



Bulk ATAC-seq
BAM / fragments



Single-cell ATAC
fragments +
annotations

1


atac-correct
Tn5 bias & SNR
95q-scaling

Pseudobulking by celltype



B cell

T/NK

Mono

2


call-footprints
Identify & score FPs

3a

ACGT
match-motifs
JASPAR / HOCOMOCO

3b

De novo motif
discovery
FASTA

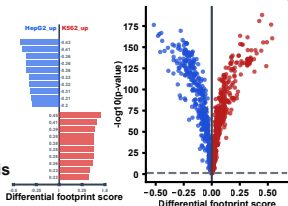
MEME / DREME
STREME

4

diff-footprints

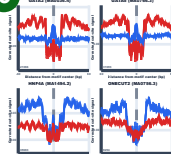


Replicate-aware
differential analysis



5

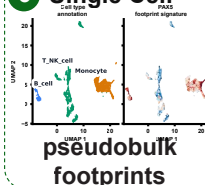
Bulk



plot-aggregate

6

Single Cell



pseudobulk
footprints